

$B$  = batch size ·  $N$  = landmark queries ·  $C$  = channels ·  $H \times W$  = spatial size

## Stage 1: Upstream spatial features

Optional upstream  
FPN-style fusion  
(final configuration)  
 $C = 384$ ;  $32 \times 32$

ScaleBlocks  $\times 2$

spatial upsampling  
 $H \times W$ :  $32 \times 32 \rightarrow 64 \times 64$   
 $\rightarrow 128 \times 128$

ScaleBlock internals  
 $2 \times 2$  transposed convolution  
stride 2  $\rightarrow$  GELU  
 $3 \times 3$  depthwise convolution  
 $\rightarrow$  2D layer normalisation

## Stage 2: Query conditioning

Group  
normalisation  
32 groups

Feature-wise linear  
modulation (FiLM)  
 $F'_q = \gamma_q \odot GN(F) + \beta_q$   
 $B \times N$  feature maps  
 $C = 384$ ;  $H \times W = 128 \times 128$

Queries  $q$   
 $N$  queries  
 $C = 384$

FiLM parameter network  
multilayer perceptron  
 $384 \rightarrow 384 \rightarrow 2 \times 384$   
ReLU · dropout 0.2  
outputs  $\gamma_q$ : scale  
and  $\beta_q$ : bias

## Stage 3: Shared spatial decoder

Shared decoder  
 $3 \times 3$  convolution  
 $C: 384 \rightarrow 128$   
ReLU  
 $3 \times 3$  convolution  
 $C: 128 \rightarrow 1$

Bilinear resize  
 $H \times W$   
 $128 \times 128 \rightarrow 64 \times 64$

Output  
 $N$  heatmaps  
 $64 \times 64$  each

Initialisation and scope  
Second convolution: weights 0, bias 0.01  
No final-head attention-mask feedback